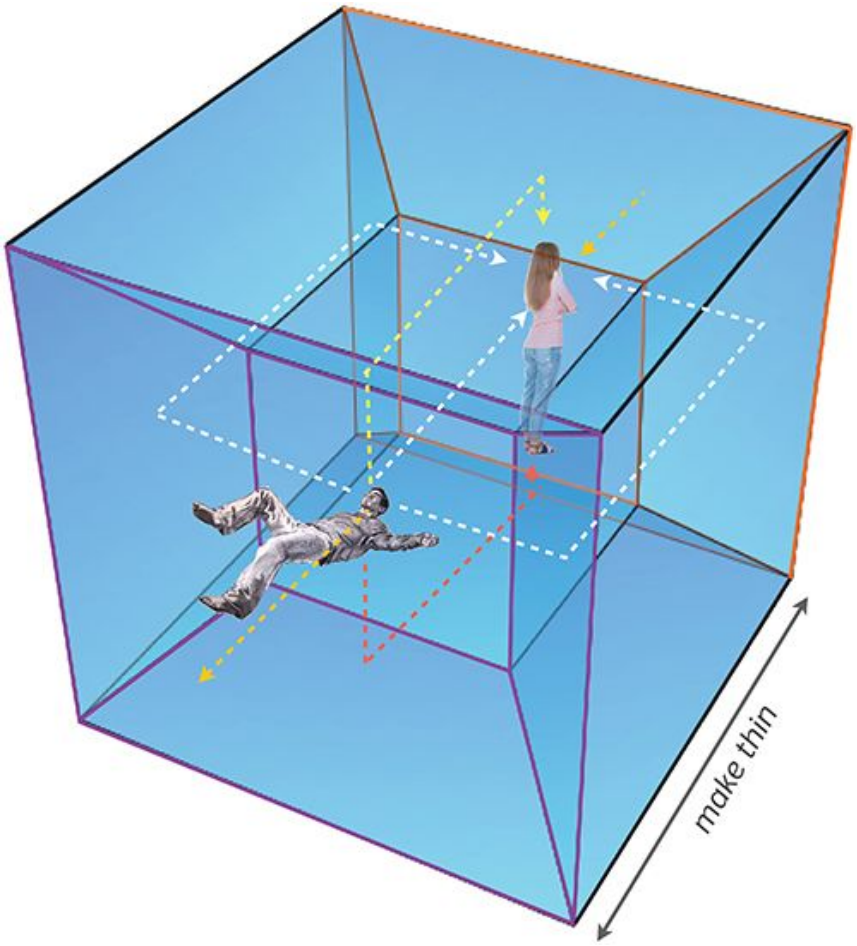


Fig. 29.6. The Cooper icon can see into Murph's bedroom (*orange edges*) by looking through each of the six walls of his face of the tesseract (*purple edges*). Here he sees an icon of Murph herself.

In Figure 29.6, all six light rays have to pass through intermediate cubes (tesseract faces) before reaching Murph's bedroom. In the movie they don't travel any noticeable distance from chamber to bedroom, so Chris and Paul must have shrunk the tesseract in one dimension; see the gray arrow and notation "make thin" in Figure 29.6.

After that shrinkage, every face of Cooper's chamber looks directly and immediately into one of the faces (wall or floor or ceiling) of Murph's bedroom with no intervening space, so to Cooper the situation looks like Figure 29.7. He sees six bedrooms, one bordering each face